Lab 4: Dead Reckoning Navigation with IMU and Magnetometer

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GitHub: [ramirez-adr] (drivers uploaded by this member)

1 Yaw Estimation

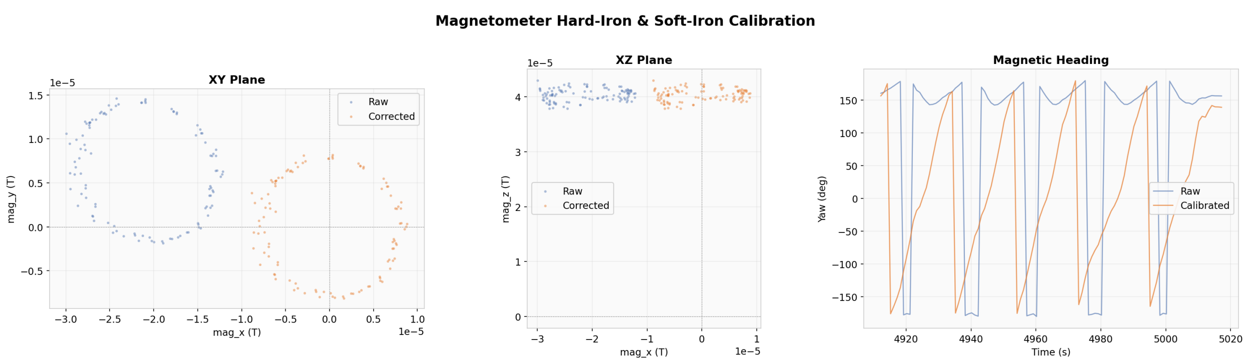

Figure 1: Magnetometer Calibration

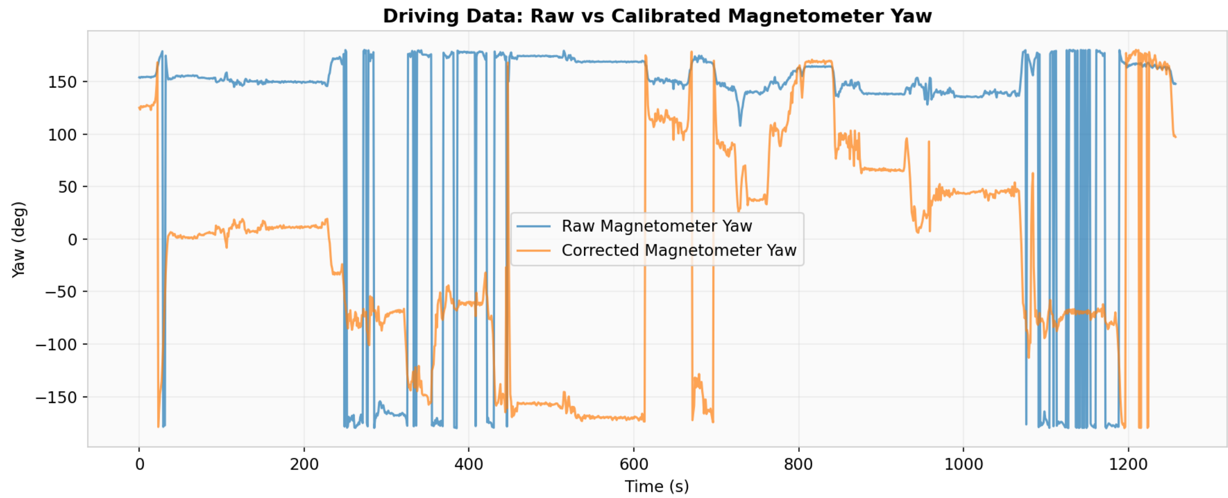

Figure 2: Driving Data Calibration

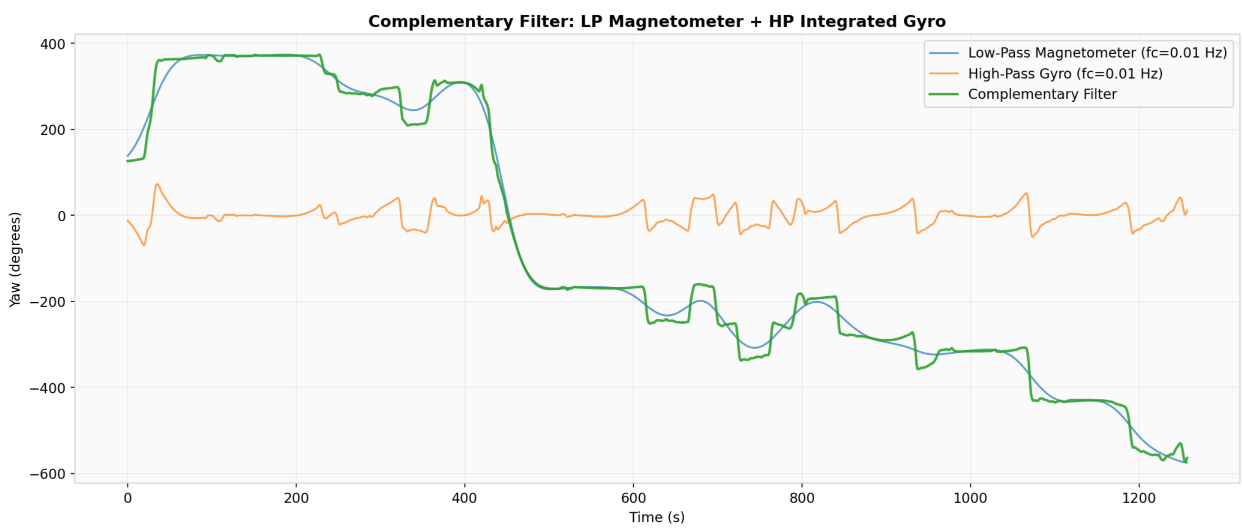

Figure 3: Complementary Filter

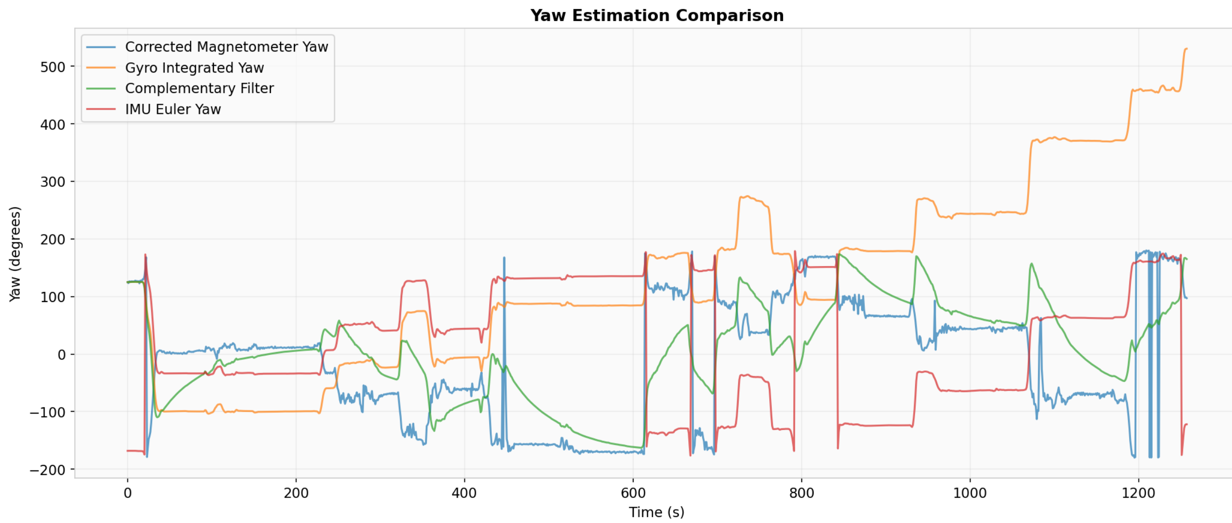

Figure 4: Yaw Estimates Comparisons

Q1: How did you calibrate the magnetometer from the data you collected? What were the sources of distortion present, and how do you know?

The magnetometer was calibrated using a 2D least-squares ellipse fit on XY magnetometer data—3D was not used because flat driving gave no Z variation, making the 3D fit unstable. For hard-iron correction, the raw XY data was centered at $(-2.09 \times 10^{-5},\ 6.35 \times 10^{-6})$ T, not the origin. This offset was subtracted and was most plausibly due to permanent magnetic sources on the vehicle. For soft-iron correction, a correction matrix from the ellipse eigendecomposition was applied to scale the axes to equal radii. This distortion is caused by ferromagnetic materials distorting the field non-uniformly. After correction, the axis ratio improved to 1.18.

Q2: How did you convert raw magnetometer readings (gauss) to angles (degrees)?

To convert raw magnetometer readings to angles, the `atan2(mag_y, mag_x)` formula was applied to the calibrated XY magnetometer readings, which returns heading in radians. The radians were then converted to degrees by multiplying by $180/\pi$. The `atan2` function was used because it resolves heading in all four quadrants, unlike `arctan` which only covers ± 90. The hard-iron and soft-iron corrections from the circle-driving calibration were applied before computing the angles.

Q3: How did you use a complementary filter to develop a combined estimate of yaw? What components of the filter were present, and what cutoff frequency(ies) did you use? Show the equation of your complementary filter.

The gyro Z angular rate was integrated using cumulative trapezoidal integration to get gyro-based yaw, and the sign was negated to match the magnetometer convention. Next, both yaw signals were unwrapped to remove ±180 discontinuities before filtering. For the filter design, 2nd-order Butterworth filters were used with a cutoff frequency of 0.01 Hz. A low-pass filter was applied to the unwrapped magnetometer yaw, and a high-pass filter was applied to the unwrapped integrated gyro yaw. The cutoff choice of 0.01 Hz means the magnetometer is trusted for changes slower than ∼100 seconds, and the gyro handles anything faster. This cutoff was chosen because the physical IMU was unable to run at 40 Hz, instead defaulting to 1 Hz, which limits the Nyquist frequency to 0.5 Hz. The complementary filter equation is:

$$\hat{\psi} = G_{LP}(s) \cdot \psi_{mag} + G_{HP}(s) \cdot \psi_{gyro} \tag{1}$$

where $G_{LP}(s) + G_{HP}(s) = 1$, with both filters being 2nd-order Butterworth at $f_c = 0.01$ Hz.

Q4: Which estimate or estimates for yaw would you trust for navigation? Why?

I would trust the complementary filter the most, as it combines the best of both sensors and corrects for drift. The magnetometer alone is too noisy for navigation, and the gyro drifts over time, as seen in Figure 4.

2 Estimation of Forward Velocity

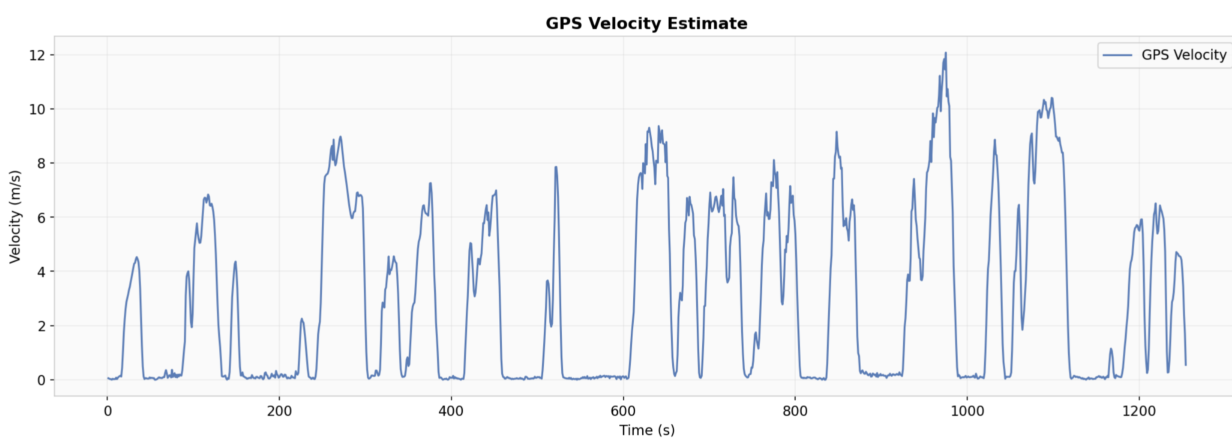

Figure 5: GPS Velocity Estimate

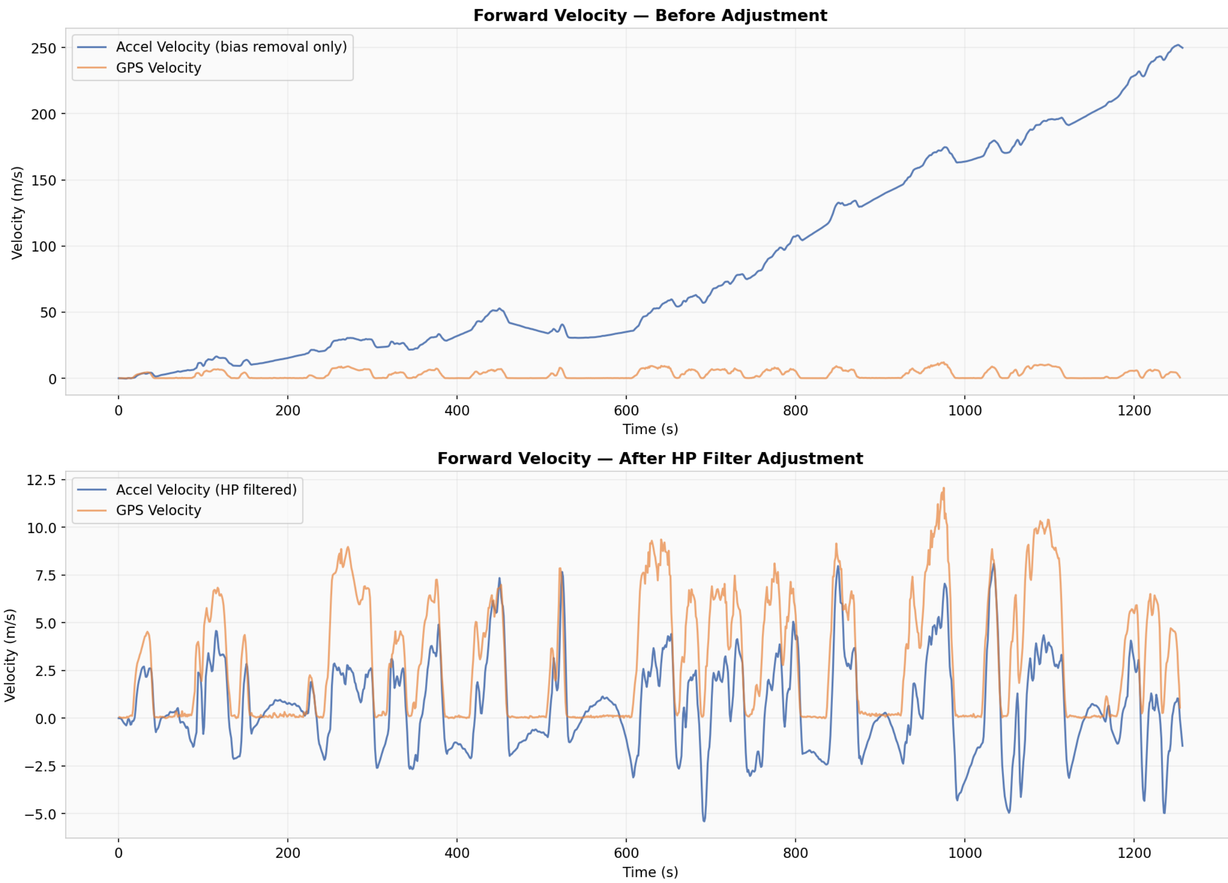

Figure 6: Forward Velocity - After Adjustment

Q5: What adjustments did you make to the forward velocity estimate, and why?

Adjustments made include first removing the accelerometer X bias estimated from the first 10 seconds of stationary data. This alone was not enough, as velocity still drifted to 250+ m/s. Therefore, a 2nd-order Butterworth high-pass filter at 0.01 Hz was applied to the debiased acceleration before integrating. This has a tradeoff, as the HP filter also attenuates real low-frequency velocity changes, so the integrated velocity underestimates sustained motion.

Q6: What discrepancies are present in the velocity estimate between accelerometer and GPS? Why?

The accelerometer velocity estimate is noisier than the GPS due to the sensor capturing road vibrations. The HP filter also introduces negative velocity values, an artifact of removing low-frequency content during filtering. The peak magnitudes also differ, with the bias-removed accelerometer velocity undershooting GPS peaks due to the HP filter attenuating sustained acceleration. The GPS estimate has its own limitations—it is computed by differencing UTM positions, so it is noisy at low speeds and can show non-zero velocity when the car is stationary due to position jitter. The most important discrepancy is that accelerometer velocity is obtained through integration, which amplifies noise and causes any residual bias to grow, whereas GPS measures position directly and does not drift.

3 Dead Reckoning

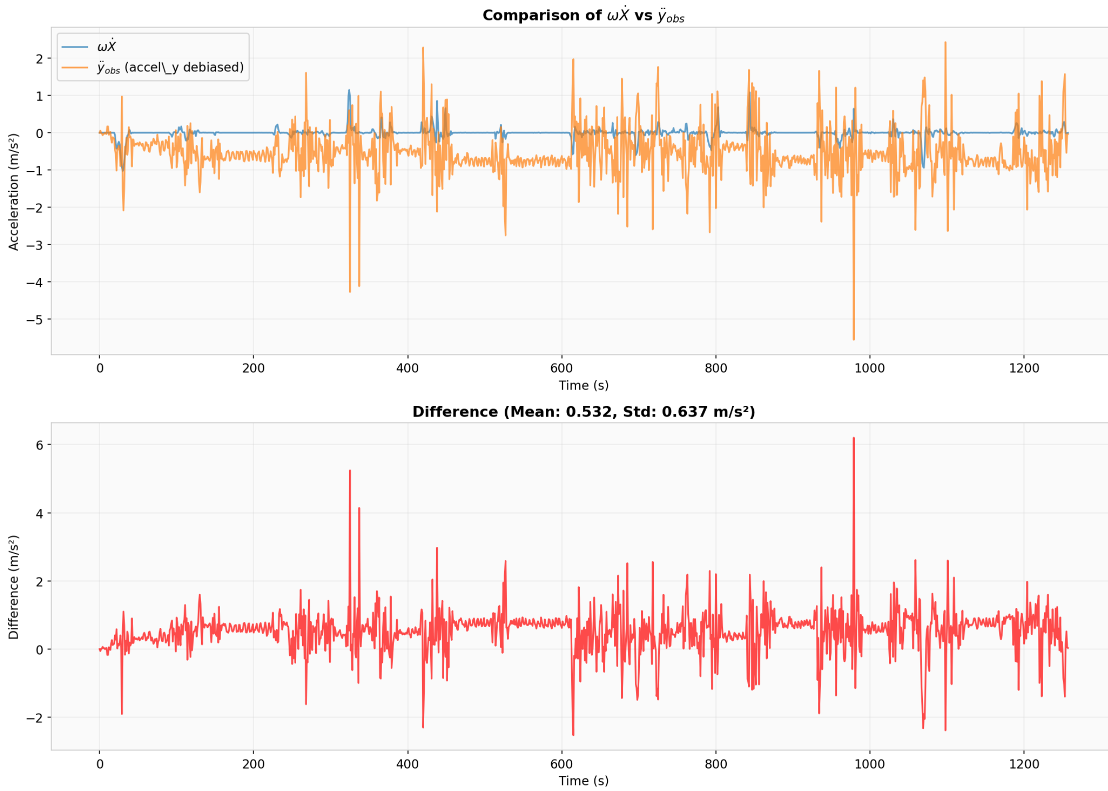

Figure 7: Comparison of $\omega\dot{X}$ and observed lateral acceleration $\ddot{y}_{obs}$.

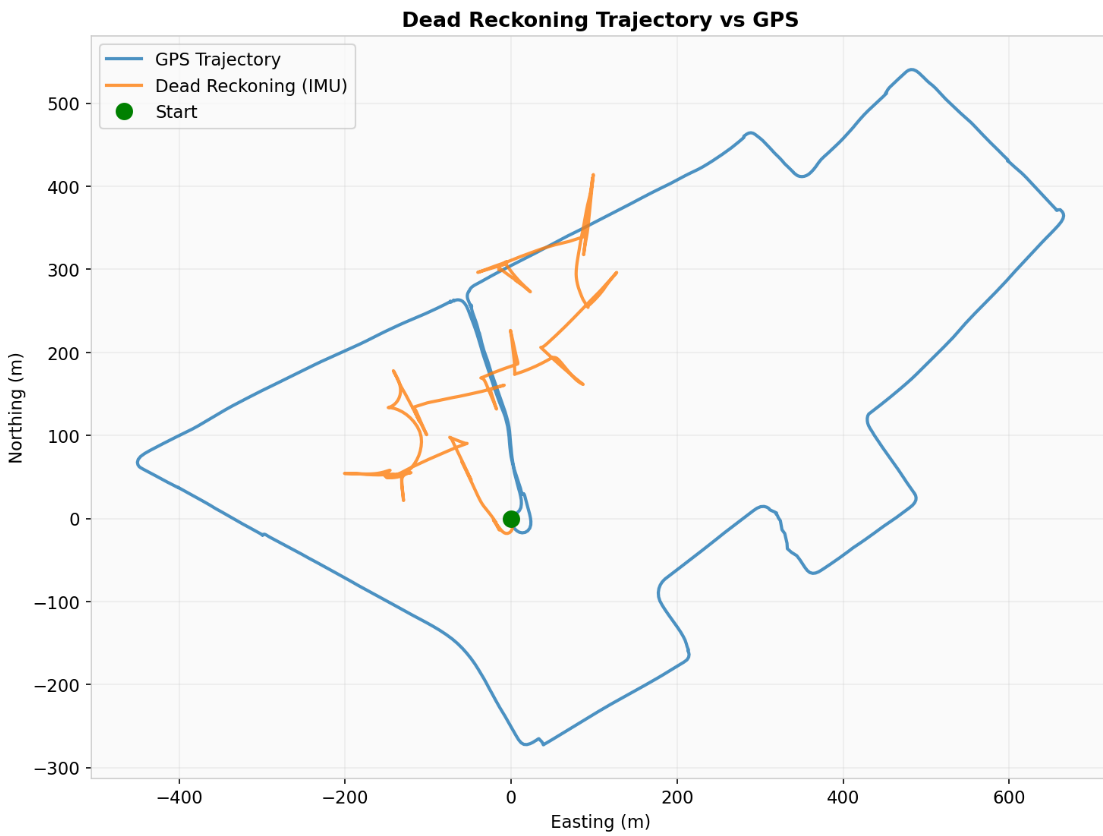

Figure 8: Dead reckoning trajectory overlaid with GPS trajectory. Starting points and initial headings aligned.

Q7: Compute $\omega\dot{X}$ and compare it to $\ddot{y}_{obs}$. How well do they agree? If there is a difference, what is it due to?

They roughly agree during turns—both show spikes at the same times, confirming the planar motion model is valid. However, $\omega\dot{X}$ is much smaller than $\ddot{y}_{obs}$ most of the time. The main reason is that the HP filter applied to the accelerometer—to prevent drift—also attenuated the real forward velocity, making $\dot{X}$ too small. In addition, there is a constant offset of approximately 0.53 m/s^2 in the difference plot. This is most plausibly due to residual a_y bias and the assumption that $x_c = 0$, when in reality the IMU was mounted on the dashboard, offset from the car's center of rotation.

Q8: Estimate the trajectory of the vehicle (x_e, x_n) from inertial data and compare with GPS. Report any scaling factor used for comparing the tracks.

The vehicle trajectory was estimated by rotating the HP-filtered forward velocity using the complementary filter heading to obtain easting and northing velocity components ($v_e = \dot{X}\sin(\psi)$, $v_n = \dot{X}\cos(\psi)$), then integrating both to get the position (x_e, x_n). A scaling factor of 1.57 was computed from the ratio of GPS total distance (3788 m) to IMU total distance

(2416 m). Without scaling, the dead reckoning trajectory stays clustered near the starting point, with a maximum position error of 815 m and only 13 seconds spent within 2 m of the GPS track. With scaling applied, the trajectory covers a more comparable area and some features match the GPS, such as the initial northwestward movement from the starting point.

The trajectory still diverges significantly due to multiple factors: the HP filter attenuates sustained velocity, causing distance to be underestimated; double integration amplifies residual bias and noise; the 1 Hz sampling rate misses fast dynamics compared to the expected 40 Hz; and cumulative heading errors compound over the 1200+ second drive.

Q9: Given the specifications of the VectorNav, how long would you expect that it is able to navigate without a position fix? For what period of time did your GPS and IMU estimates of position match closely (within 2 m)? Did the stated performance for dead reckoning match actual measurements? Why or why not?

Based on the VN-100 specifications, the gyro bias instability is approximately 10 /hr and the accelerometer bias instability is approximately 0.04 mg. Since position error from accelerometer bias grows quadratically through double integration ($\Delta x = \frac{1}{2}a \cdot t^2$), a 0.04 mg bias ($\approx 3.9 \times 10^{-4}$ m/s^2) would produce 2 m of position error in roughly:

$$t = \sqrt{\frac{2 \cdot \Delta x}{a}} = \sqrt{\frac{2 \cdot 2}{0.04 \times 10^{-3} \times 9.81}} \approx 101 \text{ seconds} \tag{2}$$

The actual performance was only 13 seconds within 2 m of GPS. This difference in performance is due to the IMU running at 1 Hz instead of 40 Hz, causing significant information loss and aliasing; the HP filter attenuating real velocity; and bias estimation from only 10 seconds of stationary data, which may not have captured the true long-term bias.